

**Predicting Student Course Completion in Online  
Learning Using Machine Learning**  
A Capstone Paper

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## **CHAPTER 1 INTRODUCTION**

### **1.1 Background of the Study**

The rapid advancement of digital technologies has transformed the landscape of education, making online learning one of the most significant innovations in contemporary teaching and learning (Dhawan, 2020; Hodges et al., 2020). Educational institutions, universities, corporate training organizations, and Massive Open Online Course (MOOC) providers increasingly rely on online learning platforms to deliver flexible, accessible, and cost-effective educational opportunities. These platforms enable learners to access instructional materials regardless of geographical location and time constraints, thereby promoting lifelong learning and widening educational access.

The global adoption of online learning accelerated significantly during the COVID-19 pandemic as educational institutions transitioned from traditional classroom instruction to virtual learning environments to ensure educational continuity (Dhawan, 2020; Hodges et al., 2020).

. Even after the pandemic, many institutions have continued integrating online learning into their educational systems because of its flexibility, scalability, and ability to support personalized learning experiences. Consequently, online learning has evolved from an alternative mode of instruction into an integral component of modern education.

Despite its flexibility and accessibility, online learning platforms continue to experience low course completion rates, with many learners failing to complete enrolled courses due to academic, motivational, and behavioral factors (Kizilcec et al., 2013; Romero & Ventura, 2020). Studies consistently report that many learners enroll in online courses but fail to complete them due to various academic, personal, technological, and motivational factors. Common reasons include inadequate time management, declining motivation, limited engagement with course materials, insufficient interaction with instructors and peers, poor internet connectivity, competing professional or family responsibilities, and a lack of prior online learning experience. These challenges not only affect learner achievement but also reduce the effectiveness and credibility of online education programs.

Educational institutions increasingly recognize that improving learner retention requires more than simply providing access to digital content. Instead, institutions must proactively identify learners who are at risk of dropping out and implement timely interventions before disengagement occurs. Traditional methods of monitoring student progress often rely on manual observation or periodic academic evaluations, which may fail to detect early warning signs. As a result, there is growing interest in applying educational data analytics and artificial intelligence to support evidence-based decision-making in online education.

Machine learning has emerged as a powerful tool for predicting educational outcomes by identifying hidden patterns within educational data and generating predictive models that support timely interventions for at-risk learners (Baker & Inventado, 2014; Romero & Ventura, 2020). By analyzing historical learner data, machine learning algorithms can identify complex relationships among demographic characteristics, academic performance, learning behaviors, and engagement indicators that influence student success. Unlike traditional statistical methods that often assume linear relationships, machine learning techniques can model complex and nonlinear interactions, making them particularly suitable for predicting educational outcomes.

Predictive analytics in education has gained considerable attention over the past decade. Previous studies have demonstrated that variables such as quiz performance, assignment completion, participation in online discussions, video viewing behavior, study habits, motivation, and previous academic achievement significantly influence course completion. Machine learning models can leverage these variables to generate predictions regarding learner success, enabling educators to implement targeted interventions such as personalized tutoring, academic advising, motivational support, or adaptive learning strategies.

However, predictive performance alone is insufficient for educational decision-making. Educational institutions require models that are not only accurate but also transparent, interpretable, and fair. Recent advances in Explainable Artificial Intelligence (XAI), including SHapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME), have made it possible to explain why machine learning models produce specific predictions. These techniques improve stakeholder trust by identifying the factors that most strongly influence prediction outcomes. Explainable Artificial Intelligence (XAI) has become increasingly important in educational applications because it enables educators to understand and interpret the reasoning behind machine learning predictions, thereby increasing transparency and trust in AI-assisted decision-making (Lundberg & Lee, 2017; Ribeiro et al., 2016).

Similarly, the increasing adoption of AI in education has raised concerns regarding fairness and bias. Machine learning models trained on historical educational data may unintentionally disadvantage certain demographic groups if underlying biases exist within the dataset. Consequently, responsible AI practices require fairness auditing using metrics such as demographic parity, equalized odds, and disparate impact to ensure that predictive models support equitable educational opportunities. Ethical Artificial Intelligence requires predictive models to operate fairly across different learner populations. Consequently, fairness assessment has become an essential component of responsible machine learning by ensuring that predictions do not systematically disadvantage specific demographic or socioeconomic groups (Mehrabi et al., 2021; Barocas et al., 2023).

This study addresses these challenges by developing and evaluating machine learning classification models to predict students' course completion status using the Online Learning Dataset 2026 (<https://www.kaggle.com/datasets/patelris/smart-learning->

[recommendation-dataset/data](#)). The study compares multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and Extreme Gradient Boosting (XGBoost), to determine the most effective predictive model. Furthermore, the selected model will be interpreted using Explainable AI techniques and evaluated through fairness analysis to ensure transparency and ethical application.

The findings of this study are expected to contribute to the growing field of educational data mining by demonstrating how machine learning can support early identification of at-risk learners, improve student retention strategies, and enhance decision-making for online learning providers. Ultimately, the proposed predictive framework seeks to promote learner success while encouraging the responsible adoption of artificial intelligence in education.

## **1.2 Problem Understanding and Framing**

### **1.2.1 Business Problem**

Online learning has become an integral component of modern education because it provides learners with flexible, accessible, and personalized learning opportunities. However, despite its widespread adoption, learner retention remains a significant challenge for educational institutions and online learning providers. Many students discontinue their courses before completion due to factors such as limited engagement, inadequate motivation, poor time management, technological barriers, and competing personal or professional commitments. High dropout rates reduce course completion rates, increase the demand for learner support services, and negatively affect institutional effectiveness and learner satisfaction.

Educational institutions require proactive approaches that can identify learners who are at risk of dropping out before disengagement occurs. Traditional monitoring methods, which rely primarily on periodic assessments and manual observation, often fail to detect early warning signs of learner withdrawal. Consequently, there is a need for a predictive decision-support system that enables educators to identify at-risk learners and implement timely interventions such as academic advising, mentoring, and personalized learning support. The central business problem addressed in this study is therefore the need to improve learner retention by accurately predicting whether a learner will successfully complete an online course or drop out.

### **1.2.2 Data Science Problem**

From a data science perspective, the problem involves developing a predictive machine learning model capable of classifying learners according to their final course outcome using historical learner data. The dataset contains demographic information, academic performance indicators, engagement metrics, technological characteristics, and personal learning attributes that collectively influence learner persistence in online education.

Although the original dataset includes three completion status categories (**Completed**, **Dropped**, and **In Progress**), this study focuses exclusively on learners whose final outcomes are known. Records labeled **In Progress** were excluded because they do not represent completed learning outcomes and would introduce uncertainty into the target variable. Consequently, the machine learning models are trained using only observations classified as **Completed** or **Dropped**, ensuring that each training instance has a definitive outcome.

The predictive model is intended not only to achieve high classification performance but also to support evidence-based educational decision-making. Accordingly, Explainable Artificial Intelligence (XAI) techniques will be employed to improve model transparency and enable educators to understand the factors influencing individual predictions.

### 1.2.3 Machine Learning Task

This study is formulated as a **supervised binary classification** problem.

The target variable is **completion\_status**, representing the learner's final course outcome. After excluding learners with an **In Progress** status, the target variable consists of two mutually exclusive classes:

- **Completed**
- **Dropped**

Binary classification is the most appropriate machine learning approach because the objective is to predict one of two predefined categorical outcomes using historical labeled data. Regression techniques are unsuitable because the target variable is categorical rather than continuous, while clustering methods are intended to discover naturally occurring groups in unlabeled data instead of predicting known outcomes.

### 1.2.4 Success Metrics

The predictive performance of the machine learning models will be evaluated using several complementary classification metrics to ensure a comprehensive assessment of model effectiveness.

**Accuracy** will measure the overall proportion of correctly classified learners. **Precision** will evaluate the proportion of learners predicted to have completed the course who actually completed it, while **Recall** will measure the model's ability to correctly identify learners who successfully completed their courses. **F1-Score**, which represents the harmonic mean of Precision and Recall, will serve as the primary technical evaluation metric because it provides a balanced assessment of model performance, particularly when class distributions are unequal. In addition, the **Area Under the Receiver Operating Characteristic Curve (ROC-AUC)** will be used to evaluate the model's ability to distinguish between learners who complete their courses and those who drop out.

across different classification thresholds. Confusion matrices and class-specific performance reports will also be analyzed to assess model performance for both learner outcomes.

## 1.2.5 Business Key Performance Indicators

The practical value of the predictive model will be assessed using business-oriented key performance indicators that reflect improvements in educational effectiveness and learner success.

The primary business objective is to **increase course completion rates** by enabling educational institutions to identify learners who may require additional academic support before they disengage from their courses. Successful implementation of the predictive model is also expected to contribute to **reduced learner dropout rates**, **improved learner retention**, and **more efficient allocation of academic support resources** by allowing educators to prioritize interventions for learners who exhibit characteristics associated with unsuccessful course completion. Ultimately, these improvements are expected to enhance learner satisfaction, strengthen institutional performance, and promote data-driven decision-making within online learning environments.

## 1.2.6 Capstone Deliverable

The primary deliverable for this phase is a clearly defined machine learning problem that aligns educational objectives with predictive analytics. Specifically, this step establishes the business problem, formulates the corresponding data science problem, identifies the machine learning task as **supervised binary classification**, defines the target variable (**completion\_status**), and specifies both the technical success metrics (Accuracy, Precision, Recall, F1-Score, and ROC-AUC) and the business key performance indicators (course completion rate, dropout rate, learner retention, resource optimization, and learner satisfaction). This problem definition serves as the foundation for the subsequent stages of data understanding, preprocessing, feature engineering, model development, and evaluation.

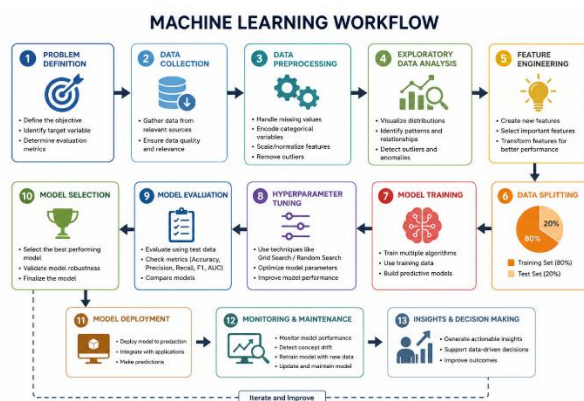


Figure 1.1 Machine Learning Workflow

## CHAPTER 2 DATA COLLECTION AND UNDERSTANDING

### 2.1 Dataset Selection and Justification

This study utilizes the **Online Learning Dataset 2026** (<https://www.kaggle.com/datasets/patelris/smart-learning-recommendation-dataset/data>), an approved custom dataset provided specifically for this capstone project. The dataset consists of learner information collected from an online learning environment and is stored in Comma-Separated Values (CSV) format, which is widely supported by data analytics and machine learning tools such as Python, Pandas, and Scikit-learn. The Online Learning Dataset 2026 was selected because it contains comprehensive learner demographic information, engagement metrics, academic performance indicators, and course outcome variables that are appropriate for educational data mining and predictive analytics (Patel, 2026).

The dataset was selected because it closely aligns with the research objective of predicting student course completion in online learning environments. It contains a comprehensive set of variables describing learners' demographic characteristics, educational background, academic performance, engagement behaviors, technological accessibility, and personal learning attributes. These variables have been identified in educational data mining literature as important predictors of student success and retention.

Unlike datasets that focus solely on academic performance, the Online Learning Dataset 2026 integrates behavioral and technological factors that reflect the multidimensional nature of online learning. This diversity enables machine learning algorithms to capture complex relationships among learner characteristics and supports the development of predictive models capable of identifying students who may require academic intervention. Educational datasets such as this provide valuable opportunities to investigate learner behavior and support data-driven interventions using machine learning techniques (Romero & Ventura, 2020).

Because the dataset contains labeled observations for the target variable (**completion\_status**), it is well suited for supervised machine learning and provides an appropriate foundation for evaluating multiple classification algorithms.

### 2.2 Dataset Overview

The Online Learning Dataset 2026 contains **4,000 learner records** and **25 variables**, comprising demographic information, academic performance indicators, learner engagement metrics, technological characteristics, and course-related information. Of the 25 variables, **13 are numerical** and **12 are categorical**, providing a balanced combination of quantitative and qualitative information for predictive modeling. An initial assessment indicates that the dataset is complete and suitable for machine learning applications. No missing values were detected during the preliminary inspection, suggesting that the dataset requires minimal data imputation. Duplicate records will be

examined during preprocessing to ensure data integrity, while continuous variables will be assessed for potential outliers using descriptive statistics, boxplots, and the Interquartile Range (IQR) method.

Table 2.1 summarizes the overall characteristics of the dataset.

| Dataset Characteristic   | Description                            |
|--------------------------|----------------------------------------|
| Dataset Name             | Online Learning Dataset 2026           |
| Source                   | Approved custom dataset                |
| File Format              | CSV                                    |
| Number of Observations   | 4,000                                  |
| Number of Variables      | 25                                     |
| Numerical Variables      | 13                                     |
| Categorical Variables    | 12                                     |
| Target Variable          | completion_status                      |
| Missing Values           | None detected                          |
| Duplicate Records        | To be verified during preprocessing    |
| Outlier Detection Method | Interquartile Range (IQR) and Boxplots |

**Table 2.1 Dataset Overview**

## 2.3 Data Quality Assessment

A preliminary data quality assessment was conducted to evaluate the completeness, consistency, and suitability of the dataset for predictive modeling.

The inspection revealed that the dataset contains **no missing values**, indicating that all observations are complete across every variable. This minimizes the need for data imputation and reduces the likelihood of introducing bias during preprocessing. Although duplicate records were not observed during the initial assessment, duplicate detection will be performed during preprocessing to ensure that each learner is represented only once in the dataset.

Continuous variables such as **age**, **hours\_per\_week**, **quiz\_scores\_avg**, **assignment\_completion\_rate**, **videos\_watched\_pct**, **course\_price\_usd**, **self\_motivation\_score**, **time\_management\_score**, and **satisfaction\_rating** will be evaluated for extreme values using boxplots and the Interquartile Range (IQR) method. Outliers identified as valid learner behavior will be retained, whereas erroneous observations resulting from data entry inconsistencies will be corrected or removed where appropriate.

The distribution of both numerical and categorical variables will be explored through descriptive statistics and graphical analysis. Histograms and boxplots will be used to examine the distributions of continuous variables, while frequency tables and bar charts will summarize categorical variables. Correlation analysis and heatmaps will also



be performed to identify relationships among numerical predictors and detect potential multicollinearity prior to model development.

## 2.4 Data Dictionary

Table 2.2 presents the variables included in the Online Learning Dataset 2026, together with their corresponding data types, measurement units or allowable values, and descriptions.

| Variable                   | Data Type   | Unit / Allowed Values                                | Description                                   |
|----------------------------|-------------|------------------------------------------------------|-----------------------------------------------|
| student_id                 | Integer     | Unique identifier                                    | Unique learner identifier                     |
| age                        | Integer     | Years                                                | Age of the learner                            |
| gender                     | Categorical | Male, Female, Other                                  | Gender of the learner                         |
| country                    | Categorical | Country names                                        | Country of residence                          |
| education_level            | Categorical | High School, Bachelor's, Master's, Doctorate, Others | Highest educational attainment                |
| course_category            | Categorical | Subject categories                                   | Category of the enrolled course               |
| platform                   | Categorical | Platform names                                       | Online learning platform                      |
| course_duration_weeks      | Integer     | Weeks                                                | Duration of the course                        |
| hours_per_week             | Integer     | Hours                                                | Average weekly study hours                    |
| quiz_scores_avg            | Float       | 0–100 (%)                                            | Average quiz score                            |
| assignment_completion_rate | Float       | 0–100 (%)                                            | Percentage of assignments completed           |
| forum_posts                | Integer     | Count                                                | Number of forum contributions                 |
| videos_watched_pct         | Float       | 0–100 (%)                                            | Percentage of learning videos viewed          |
| device                     | Categorical | Desktop, Laptop, Tablet, Mobile                      | Primary learning device                       |
| internet_speed             | Categorical | Slow, Moderate, Fast                                 | Internet connection quality                   |
| previous_courses_completed | Integer     | Count                                                | Number of previously completed online courses |
| financial_aid              | Categorical | Yes, No                                              | Financial aid status                          |
| course_price_usd           | Float       | US Dollars                                           | Cost of the course                            |
| self_motivation_score      | Integer     | 1–10                                                 | Self-rated motivation level                   |
| time_management_score      | Integer     | 1–10                                                 | Self-rated time management ability            |
| learning_disability        | Categorical | Yes, No                                              | Presence of a reported learning disability    |

|                     |             |                                           |                              |
|---------------------|-------------|-------------------------------------------|------------------------------|
| employment_status   | Categorical | Full-time, Part-time, Unemployed, Student | Current employment status    |
| completion_status   | Categorical | Completed, In Progress, Dropped           | <b>Target variable</b>       |
| final_grade         | Categorical | A, B, C, D, F                             | Final academic grade*        |
| satisfaction_rating | Float       | 1–5                                       | Overall learner satisfaction |

\*The variable **final\_grade** will be excluded from predictive model training if it is determined to be available only after course completion, as including it could introduce target leakage and artificially inflate model performance.

**Table 2.2 Data Dictionary**

## 2.5 Summary

The Online Learning Dataset 2026 provides a high-quality foundation for developing predictive machine learning models. Its combination of demographic, academic, behavioral, engagement, and technological variables enables comprehensive analysis of factors influencing student course completion. The absence of missing values, together with the balanced representation of numerical and categorical features, minimizes preprocessing complexity while supporting robust model development. The preliminary assessment confirms that the dataset is suitable for supervised multiclass classification and provides sufficient information for exploratory data analysis, feature engineering, explainable artificial intelligence, and fairness evaluation in subsequent phases of the study.

## CHAPTER 3 DATA PREPROCESSING, APPLIED EXPLORATORY DATA ANALYSIS, AND FEATURE ENGINEERING

### 3.1 Data Preprocessing

Data preprocessing is a critical phase of the machine learning pipeline that aims to improve data quality and prepare the dataset for predictive modeling. Raw datasets often contain inconsistencies, missing values, duplicate records, outliers, and variables measured on different scales, all of which can adversely affect the performance and reliability of machine learning algorithms. Effective preprocessing ensures that the data are accurate, consistent, and suitable for analysis, thereby improving model performance, reducing computational complexity, and enhancing the validity of the study's findings. Data preprocessing is widely recognized as one of the most critical stages in the machine learning pipeline because the quality of the input data directly influences model performance (Han et al., 2012; Géron, 2022).

In this study, the **Online Learning Dataset 2026** was subjected to a systematic preprocessing procedure before exploratory data analysis and model development. The preprocessing activities included data cleaning, feature transformation, feature engineering, feature selection, and dimensionality reduction. These procedures were implemented to ensure that the dataset accurately represented learner characteristics and could effectively support the development of robust machine learning models for predicting course completion status.

The first stage of preprocessing involved evaluating the quality of the dataset by examining missing values, duplicate records, and potential outliers. Missing values were assessed to determine whether data imputation was necessary, while duplicate records were identified and removed to prevent redundant observations from influencing model training. Continuous numerical variables were also examined for extreme values using descriptive statistics and the Interquartile Range (IQR) method. Rather than automatically removing outliers, each observation was evaluated to determine whether it represented a valid learner characteristic or a possible data entry error. This approach ensured that meaningful variations in learner behavior were preserved while minimizing the influence of erroneous data.

Following data cleaning, feature transformation techniques were applied to prepare the variables for machine learning algorithms. Categorical variables were converted into numerical representations using appropriate encoding methods, while continuous numerical variables were standardized to ensure that differences in measurement scales did not disproportionately influence model training. In addition, feature engineering techniques were employed to create new variables that better captured learner engagement and academic performance by combining related attributes into more informative indicators.

To further improve model efficiency and interpretability, feature selection techniques were implemented to identify the most informative predictors while reducing

redundancy among variables. Dimensionality reduction using Principal Component Analysis (PCA) was also considered to transform correlated variables into a smaller set of principal components while retaining most of the information contained in the original dataset. Finally, Explainable Artificial Intelligence (XAI) techniques were incorporated to identify the variables that contributed most significantly to the model's predictions and to support transparent and responsible use of machine learning in educational decision-making.

Through these preprocessing procedures, the dataset was transformed into a clean, consistent, and analytically suitable form that supports reliable exploratory data analysis, robust predictive modeling, and meaningful interpretation of the factors influencing student course completion.

| Attribute                           | Description                                                           |
|-------------------------------------|-----------------------------------------------------------------------|
| <b>Dataset Name</b>                 | Online Learning Dataset 2026                                          |
| <b>Source</b>                       | Synthetic dataset for educational machine learning analysis           |
| <b>Number of Records</b>            | 4,000 learner records                                                 |
| <b>Number of Variables</b>          | 25 variables                                                          |
| <b>Target Variable</b>              | completion_status                                                     |
| <b>Prediction Task</b>              | Binary Classification (Completed vs. Dropped)                         |
| <b>Original Target Classes</b>      | Completed, Dropped, In Progress                                       |
| <b>Final Target Classes</b>         | Completed, Dropped                                                    |
| <b>Numerical Variables</b>          | 13                                                                    |
| <b>Categorical Variables</b>        | 12                                                                    |
| <b>Missing Values</b>               | None (100% data completeness)                                         |
| <b>Duplicate Records</b>            | None                                                                  |
| <b>Machine Learning Environment</b> | Google Colab (Python)                                                 |
| <b>Primary Libraries</b>            | Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost, SHAP, LIME |

**Table 3.1 Dataset Information Before Preprocessing**

## 3.2 Data Quality Assessment

The initial assessment confirmed that the dataset consisted of learner demographic information, course participation metrics, academic performance indicators, and technological factors associated with online learning. Before model development, the dataset was evaluated to ensure its quality and suitability for predictive analysis.

### 3.2.1 Missing Values and Data Completeness

The missing value analysis revealed that all variables contained complete observations, resulting in a data completeness rate of 100 percent. No missing values were detected across the 4,000 learner records; therefore, no data imputation techniques were required. The complete dataset ensured that all observations could be retained

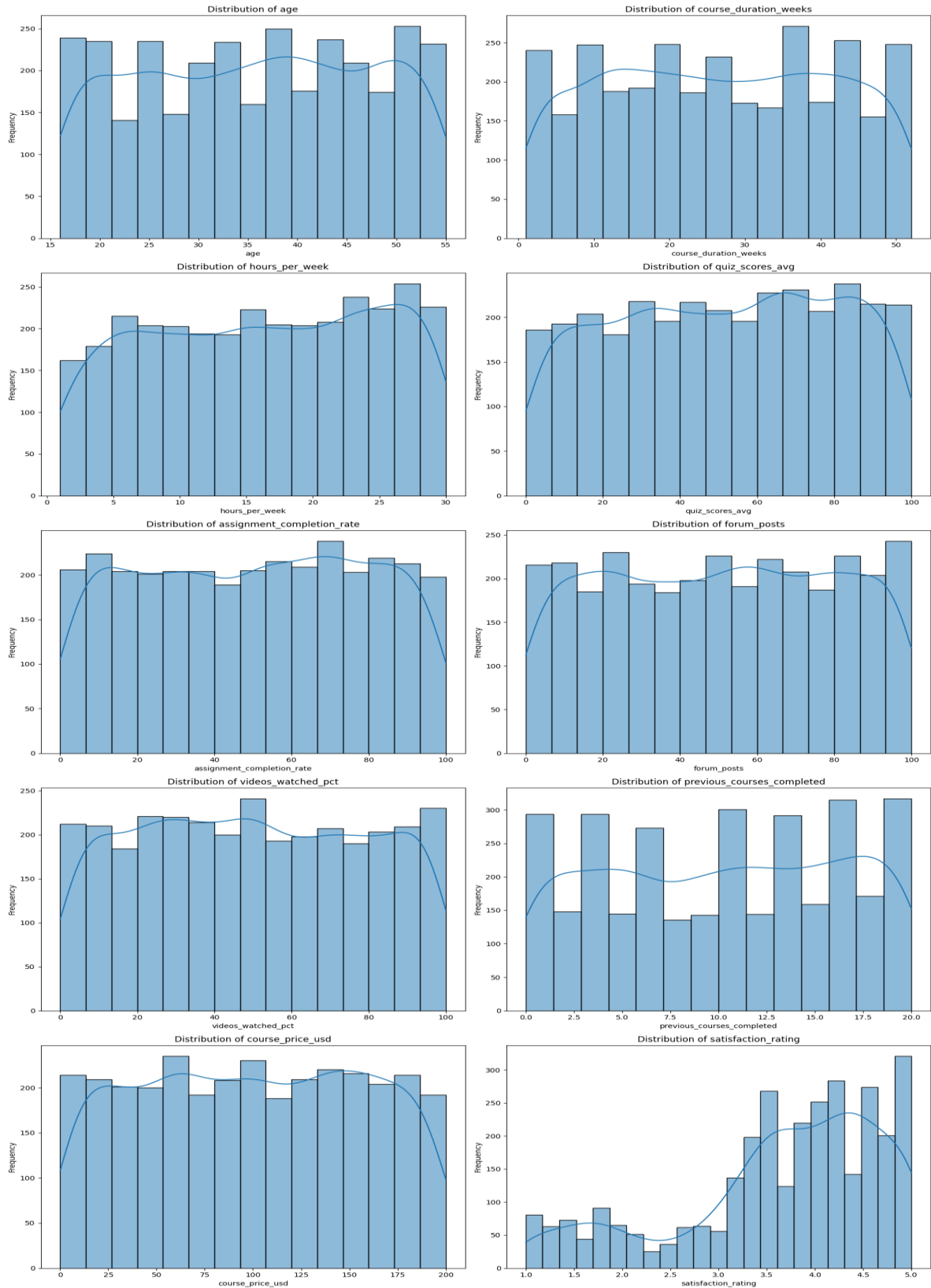
during preprocessing, providing a reliable foundation for machine learning model development.

### 3.2.2 Descriptive Statistics

Descriptive statistics were computed to summarize the central tendency, variability, and distribution of the numerical variables included in the Online Learning Dataset 2026. Statistical measures such as the minimum, maximum, mean, and standard deviation were generated to provide an overview of learner demographics, engagement, academic performance, and other relevant characteristics. These statistics offer an initial understanding of the dataset and help identify patterns, trends, and potential anomalies that may influence the predictive modeling process.

The summary statistics indicate substantial variation across several learner-related variables, reflecting the diversity of the online learning population. For example, learner age spans a wide range, while variables such as **hours per week**, **quiz scores**, **assignment completion rate**, and **videos watched percentage** exhibit differing levels of engagement and academic performance among learners. Likewise, the distributions of **self-motivation** and **time management** scores suggest varying levels of learner preparedness and study habits. Overall, the descriptive statistics confirm that the dataset captures a broad spectrum of learner characteristics, making it suitable for developing robust machine learning models.

To further examine the characteristics of the dataset, histograms were generated for both numerical and categorical variables. The histograms of numerical variables illustrate the distribution, spread, and skewness of continuous features, while the categorical variable histograms present the frequency distribution of each category. These visualizations complement the descriptive statistics by providing a clearer understanding of the data distributions and supporting the identification of potential patterns that may influence learner course completion.

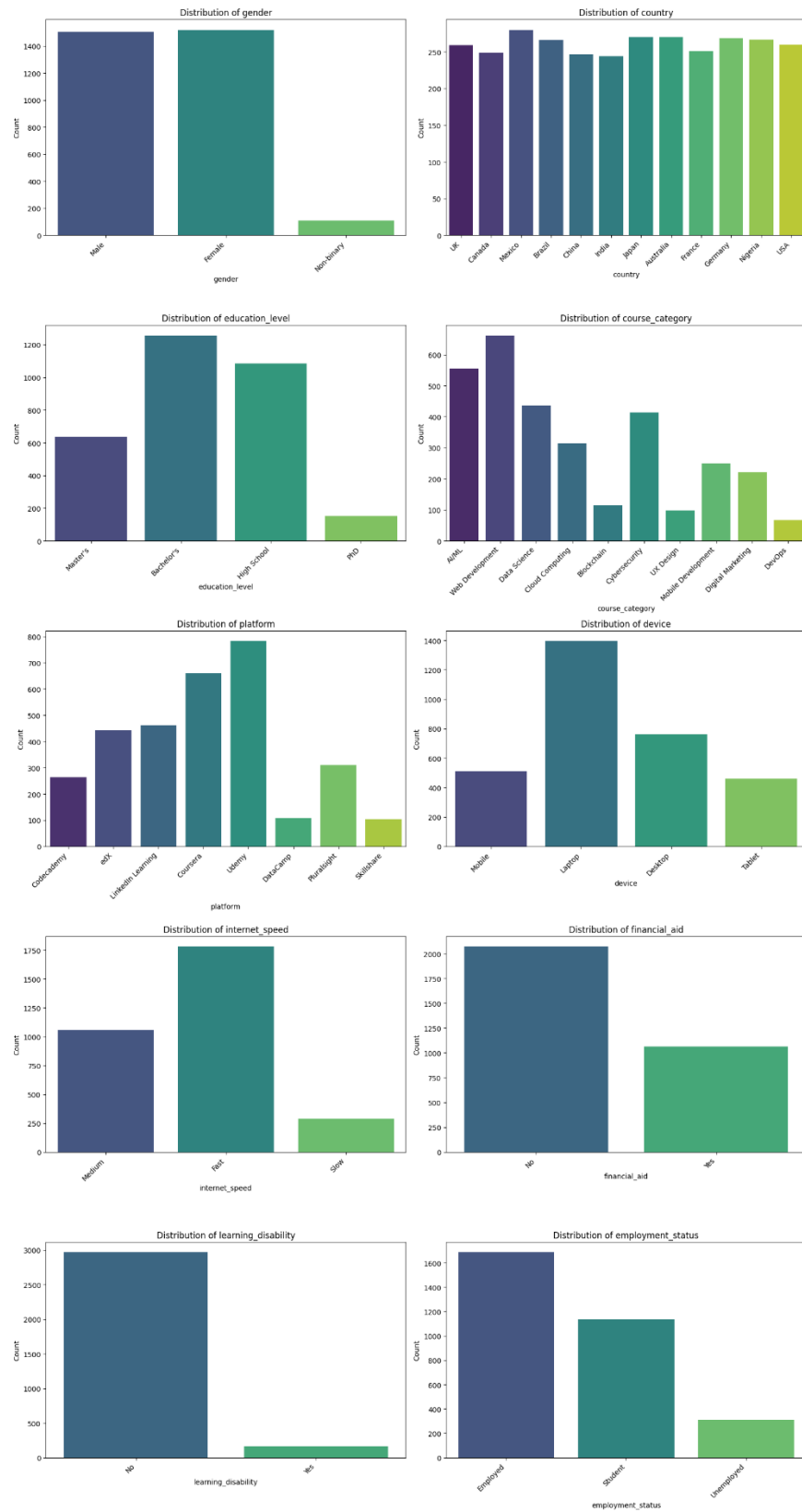


**Figure 3.1 Histograms for Numerical Variables**

## Summary of Histogram Observations

Based on the generated histograms for the numerical continuous features, here are the key observations regarding their distributions:

- **Age:** The distribution appears relatively uniform with slight peaks around typical age groups for higher education or professional development. It doesn't show strong skewness, indicating a fairly even spread of ages.
- **Course Duration Weeks:** Similar to age, the course duration seems to be spread out relatively evenly across its range, suggesting a variety of course lengths are available and taken. No strong skew or prominent modes were observed.
- **Hours Per Week:** This also exhibits a relatively uniform distribution, implying that students commit varying amounts of time to their studies each week, without strong clustering at particular hourly commitments.
- **Quiz Scores Avg, Assignment Completion Rate, Videos Watched Pct, Forum Posts:** These engagement and performance metrics generally show distributions that are somewhat uniform across their range, with some slight variations. There isn't a strong concentration at either very high or very low ends for most students, indicating a diverse range of performance and engagement levels. This suggests that students exhibit varied levels of effort and success.
- **Previous Courses Completed:** The distribution for the number of previous courses completed is also quite spread out, indicating a mix of new learners and those with significant prior online learning experience.
- **Course Price USD:** The course prices are fairly uniformly distributed across the range, suggesting that courses are offered at a variety of price points.
- **Satisfaction Rating:** This feature shows a slightly skewed distribution towards higher satisfaction ratings, with a notable peak around the 4.0-4.5 mark. There is also a smaller peak around the 1.0-1.5 mark, which corresponds to the outliers identified earlier, representing students with very low satisfaction. This bimodal tendency, though subtle, suggests two groups: generally satisfied students and a smaller group of highly dissatisfied students.



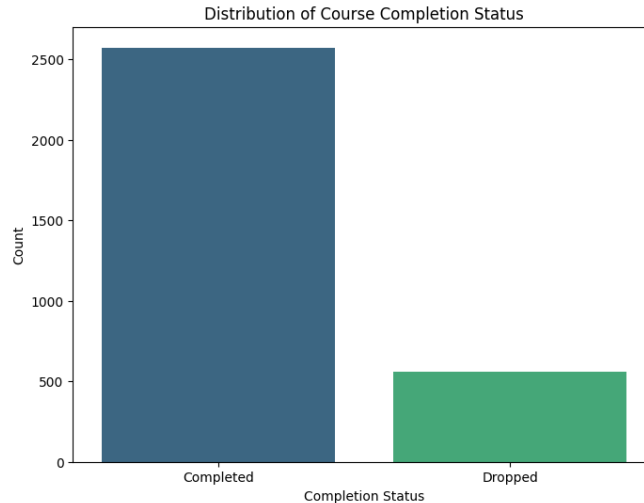
**Figure 3.2 Count Plots for Categorical Variables**



## Summary of Count Plot Observations

Based on the generated count plots for the categorical features, here are the key observations regarding their distributions:

- **Gender:** There is a relatively balanced distribution between 'Male' and 'Female' students, with 'Non-binary' representing a much smaller proportion. This indicates that gender diversity is present in the dataset, with a slight skew towards 'Female'.
- **Country:** Students come from a diverse set of countries, with 'Mexico' and 'USA' showing slightly higher counts, followed by 'India', 'Canada', 'UK', and 'Germany'. There isn't a single dominant country, suggesting a global reach for the online learning platform.
- **Education Level:** 'Bachelor's' degree holders form the largest group, followed by 'Master's', 'High School', and 'PhD' students. This suggests that the online courses are popular among individuals seeking to enhance their undergraduate or postgraduate qualifications.
- **Course Category:** 'Web Development', 'AI/ML', and 'Data Science' are the most popular course categories, indicating a high demand for tech-related skills. 'Blockchain' and 'Mobile Development' have fewer enrollments.
- **Platform:** 'Udemy' appears to be the most popular learning platform, followed by 'Coursera' and 'edX'. This highlights the preference of students for these well-established platforms.
- **Device:** 'Laptop' is the overwhelmingly dominant device used for online learning, followed by 'Mobile' and 'Desktop'. 'Tablet' use is minimal, which is expected as laptops offer a better experience for most online learning activities.
- **Internet Speed:** 'Fast' internet speed is reported by the majority of students, followed by 'Moderate' and 'Slow'. This is a positive sign for online learning, as good internet connectivity is crucial for a smooth learning experience.
- **Financial Aid:** A significant majority of students (No) do not receive financial aid, while a substantial portion (Yes) do. This implies that while some students are self-funding, financial assistance is still a notable factor for many.
- **Learning Disability:** Most students report having 'No' learning disability, with a very small percentage reporting 'Yes'. This indicates that the dataset primarily represents students without diagnosed learning disabilities.
- **Employment Status:** 'Employed' students form the largest group, followed by 'Student' and 'Unemployed'. This suggests that online learning is widely adopted by working professionals and traditional students, possibly for career advancement or supplementary education.



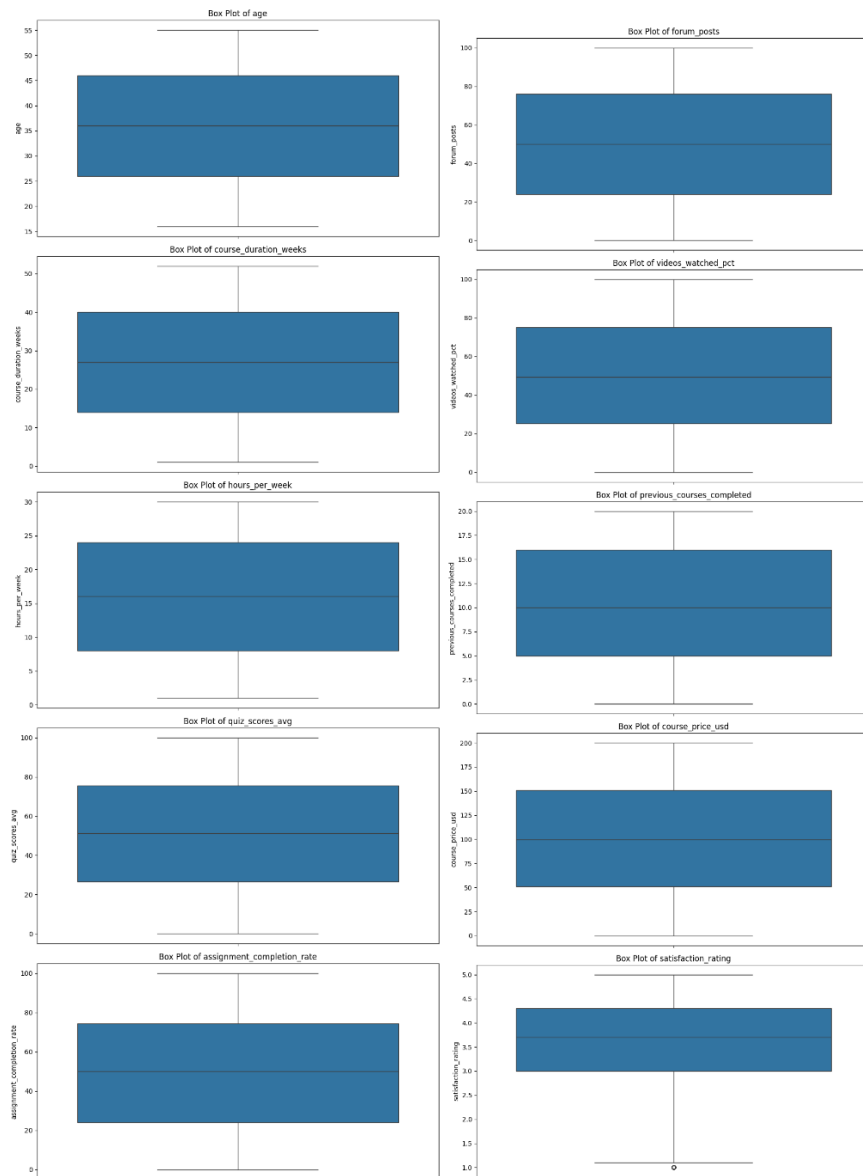
**Figure 3.3 Target Distribution Plot**

#### Summary of Target Distribution Plot Observations

The count plot for completion\_status clearly illustrates the distribution of the target variable. Key observations include:

- **Class Imbalance:** There is a significant class imbalance between the 'Completed' and 'Dropped' categories. The 'Completed' class has a much higher count (approximately 2500 students) compared to the 'Dropped' class (approximately 500 students).
- **Dominant Class:** The 'Completed' class is the dominant class, representing the majority of students in the dataset.
- **Implications:** This imbalance is a crucial factor to consider for subsequent machine learning model development. Models trained on imbalanced datasets can be biased towards the majority class, leading to poor predictive performance on the minority class ('Dropped' students in this case). This will require careful handling during model training and evaluation, potentially using techniques like resampling (oversampling/undersampling) or specialized evaluation metrics (e.g., F1-score, ROC-AUC) that are less sensitive to class imbalance.

### 3.2.4 Boxplot Analysis



**Figure 3.4 Initial Boxplot Analysis**

Boxplots were generated for the numerical variables to provide an initial assessment of their distributions and identify potential extreme values. The plots indicate that while several variables exhibit observations beyond the interquartile range (IQR), these values appear to represent natural variations in learner characteristics rather than obvious data entry errors. The boxplots also show the spread, central tendency, and variability of each numerical feature, providing valuable insights that guided the subsequent outlier assessment and preprocessing decisions.

### 3.2.3 Data Dictionary

A data dictionary was prepared to provide a comprehensive description of each variable included in the dataset. The dictionary specifies the variable name, data type, allowable values or measurement units, and a brief description of each feature. This serves as a reference throughout the preprocessing and model development stages.

| Variable                   | Data Type   | Unit / Allowed Values                                | Description                                   |
|----------------------------|-------------|------------------------------------------------------|-----------------------------------------------|
| student_id                 | Integer     | Unique identifier                                    | Unique learner identifier                     |
| age                        | Integer     | Years                                                | Age of the learner                            |
| gender                     | Categorical | Male, Female, Other                                  | Gender of the learner                         |
| country                    | Categorical | Country names                                        | Country of residence                          |
| education_level            | Categorical | High School, Bachelor's, Master's, Doctorate, Others | Highest educational attainment                |
| course_category            | Categorical | Subject categories                                   | Category of the enrolled course               |
| platform                   | Categorical | Platform names                                       | Online learning platform                      |
| course_duration_weeks      | Integer     | Weeks                                                | Duration of the course                        |
| hours_per_week             | Integer     | Hours                                                | Average weekly study hours                    |
| quiz_scores_avg            | Float       | 0–100 (%)                                            | Average quiz score                            |
| assignment_completion_rate | Float       | 0–100 (%)                                            | Percentage of assignments completed           |
| forum_posts                | Integer     | Count                                                | Number of forum contributions                 |
| videos_watched_pct         | Float       | 0–100 (%)                                            | Percentage of learning videos viewed          |
| device                     | Categorical | Desktop, Laptop, Tablet, Mobile                      | Primary learning device                       |
| internet_speed             | Categorical | Slow, Moderate, Fast                                 | Internet connection quality                   |
| previous_courses_completed | Integer     | Count                                                | Number of previously completed online courses |
| financial_aid              | Categorical | Yes, No                                              | Financial aid status                          |
| course_price_usd           | Float       | US Dollars                                           | Cost of the course                            |
| self_motivation_score      | Integer     | 1–10                                                 | Self-rated motivation level                   |
| time_management_score      | Integer     | 1–10                                                 | Self-rated time management ability            |
| learning_disability        | Categorical | Yes, No                                              | Presence of a reported learning disability    |

|                     |             |                                           |                              |
|---------------------|-------------|-------------------------------------------|------------------------------|
| employment_status   | Categorical | Full-time, Part-time, Unemployed, Student | Current employment status    |
| completion_status   | Categorical | Completed, In Progress, Dropped           | <b>Target variable</b>       |
| final_grade         | Categorical | A, B, C, D, F                             | Final academic grade*        |
| satisfaction_rating | Float       | 1–5                                       | Overall learner satisfaction |

**Table 3.1 Data Dictionary**

### 3.3 Data Cleaning

#### 3.3.1 Feature and Record Selection

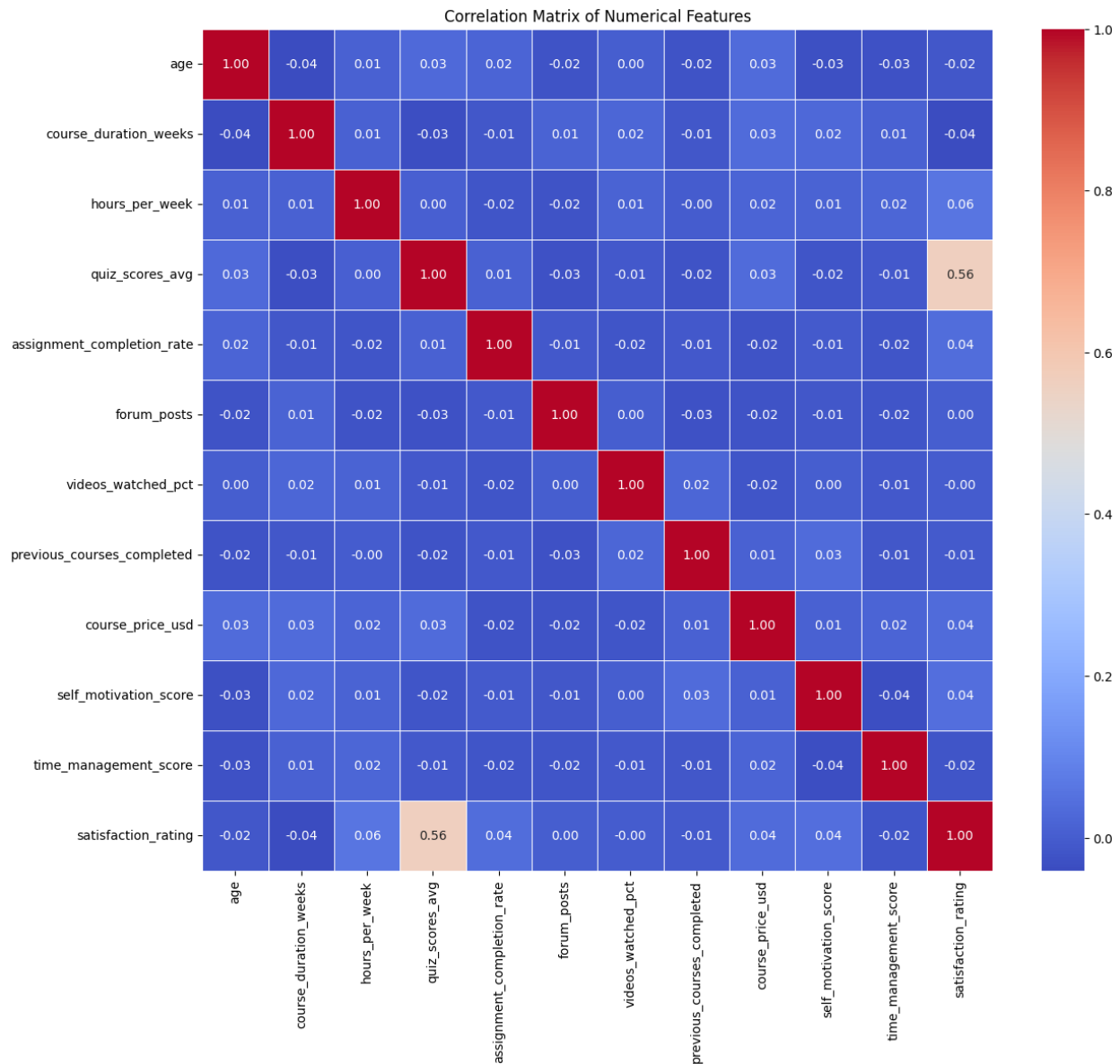
Prior to model development, selected variables and records were removed to ensure the integrity and validity of the predictive analysis. The **student\_id** variable was excluded because it serves only as a unique identifier and does not provide meaningful predictive information. Records with a **completion\_status** of "In Progress" were also removed since they do not represent final learning outcomes and could introduce uncertainty into the target variable. Furthermore, the **final\_grade** variable was excluded from the feature set because it is determined only after course completion and would introduce **target leakage** if used as a predictor. Removing these variables and records ensures that the machine learning models are trained using only relevant information that would realistically be available when predicting learner outcomes.

#### 3.3.2 Detection of Outlier Using IQR

Outlier detection was conducted using the Interquartile Range (IQR) method to identify extreme observations within the numerical variables. The analysis revealed that **most numerical continuous variables contained no significant outliers**, indicating that the dataset exhibits generally consistent distributions suitable for machine learning analysis. The only variable identified with outliers was **satisfaction\_rating**, which contained **111 outlier observations**, representing approximately **3.54%** of the dataset. These outliers were observed at the lower end of the satisfaction scale.

Further examination indicated that these observations represent legitimate learner responses rather than data entry errors or anomalies. Low satisfaction ratings are meaningful indicators of learner experience and may provide valuable information for predicting course completion outcomes. Consequently, the identified outliers were **retained** in the dataset to preserve the natural variability of learner satisfaction and to enable the machine learning models to learn from authentic patterns present in the data. Retaining these observations enhances the practical applicability and generalizability of the predictive models in real-world online learning environments.

### 3.4 Exploratory Data Analysis



**Figure 3.5 Correlation Heatmap**

The correlation matrix illustrates the relationships among numerical variables within the dataset. Most variables exhibit weak to moderate correlations, indicating limited multicollinearity. Stronger positive relationships were observed among learner engagement variables such as assignment completion rate, quiz scores, and videos watched, suggesting that increased engagement is associated with improved academic performance. These findings support the inclusion of composite engagement measures during feature engineering.

Upon examining the generated correlation matrix and heatmap, the following observations can be made regarding strong correlations and potential multicollinearity:

1. **Absence of Very Strong Correlations:** There are no pairs of numerical features with correlation coefficients significantly higher than 0.7 or lower than -0.7. This indicates that, based on a common threshold for strong correlation, there are no immediate concerns about severe multicollinearity among the numerical predictors in this dataset.
2. **Moderate Correlation:** The most notable correlation is a positive one of approximately **0.56** between `quiz_scores_avg` and `satisfaction_rating`. This suggests that students who perform better on quizzes tend to report higher satisfaction with the course. While this is a moderate relationship, it is not strong enough to cause significant multicollinearity issues for most machine learning models.
3. **Weak Correlations:** All other numerical features exhibit very weak linear correlations with each other, with most coefficients being close to zero. This implies that these features are largely independent in a linear sense.

### Implications for Model Building:

- **Multicollinearity:** Given the absence of very strong correlations (e.g.,  $>0.7$  or  $<-0.7$ ), multicollinearity is unlikely to be a major concern for this dataset in its current state. This is beneficial because high multicollinearity can destabilize model coefficients (especially in linear models like Logistic Regression), make interpretation difficult, and reduce the statistical significance of individual predictors.
- **Feature Redundancy:** Since no two features are highly correlated, it is unlikely that any single numerical feature is redundant due to information overlap with another. This means that, from a linear perspective, most numerical features bring unique information to the model.
- **Model Performance:** The moderate correlation between `quiz_scores_avg` and `satisfaction_rating` is interesting and suggests that both could be important predictors. However, this level of correlation is generally acceptable and may not require special handling like feature removal or combination, especially for robust models like tree-based algorithms (e.g., Random Forest, XGBoost) which are less sensitive to multicollinearity than linear models.

## 3.5 Feature Engineering

Feature engineering was performed to transform the original dataset into a format suitable for machine learning while enhancing its predictive capability. Categorical variables were converted into numerical representations using appropriate encoding techniques to enable processing by classification algorithms. Numerical features were standardized through feature scaling to ensure that variables with different measurement

ranges contributed equally during model training, particularly for algorithms such as Logistic Regression and Support Vector Machine. In addition, domain-derived features were created by combining related variables to better represent learner engagement and academic performance. These feature engineering techniques improve data representation, reduce potential bias caused by differing feature scales, and enhance the overall performance and interpretability of the predictive models.

Feature engineering transforms raw variables into more informative representations that improve predictive performance and model interpretability (Kuhn & Johnson, 2019).

### 3.5.1 Categorical Encoding

Categorical encoding was performed to transform non-numerical variables into numerical representations suitable for machine learning algorithms. Since most classification models require numerical input, categorical features were converted while preserving the information contained within each category.

The predictor variables were encoded using **One-Hot Encoding**, which creates separate binary indicator variables for each category of a categorical feature. To minimize multicollinearity and reduce redundant information, the first category of each encoded variable was dropped by setting the `drop_first=True` parameter during the encoding process. This approach enables the machine learning models to process nominal categorical variables without introducing artificial ordinal relationships between categories.

The target variable, **completion\_status**, was converted into a binary numerical format by mapping **Completed** to **1** and **Dropped** to **0**. Records with a status of **In Progress** were removed prior to encoding because they do not represent final learner outcomes and were therefore not suitable for the binary classification task.

The application of One-Hot Encoding and binary target encoding produced a machine learning-ready dataset while preserving the original information contained in the categorical variables. This preprocessing step ensures compatibility with the classification algorithms implemented in the subsequent stages of the study.

### 3.5.2 Feature Scaling

Feature scaling standardized numerical variables to a common scale. This process prevents variables with larger numerical ranges from dominating the learning process and improves the convergence of algorithms such as Logistic Regression and Support Vector Machines.



Standardization ensures that variables measured on different scales contribute equally during model training, particularly for algorithms that rely on distance-based optimization (Géron, 2022).

### 3.5.3 Domain-Derived Features

Three domain-derived features were engineered to better represent learner behavior and academic performance. The Engagement Score summarizes learner participation by combining assignment completion rate, video viewing percentage, and forum activity. The Academic Index integrates quiz scores and assignment completion rates to provide an overall measure of academic achievement. Finally, the Learning Experience Index combines previous course completion and learner satisfaction to represent prior online learning experience. These engineered variables provide higher-level representations of learner characteristics that enhance predictive performance.

### 3.6 Feature Selection

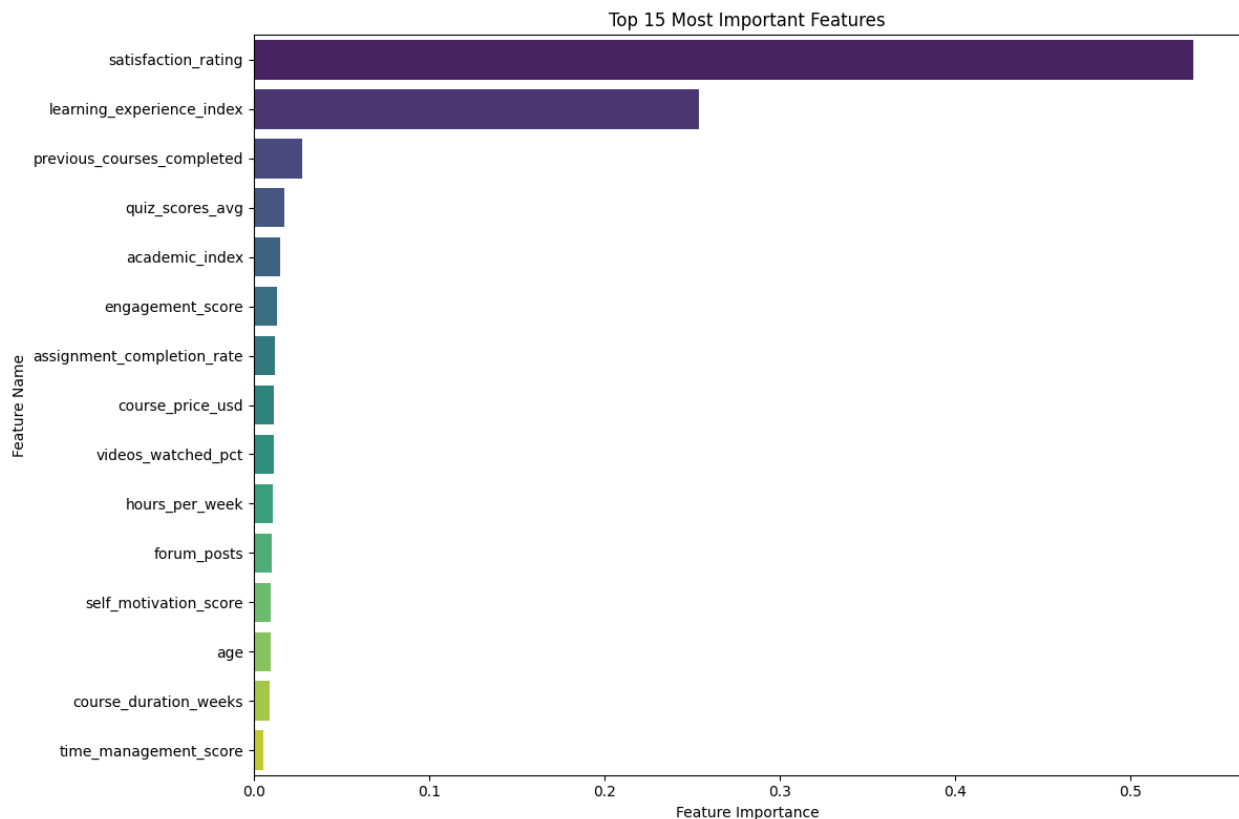


Figure 3.6 Feature Importance

Feature selection identified the variables that contributed most significantly to predicting course completion. Variables associated with learner engagement, academic

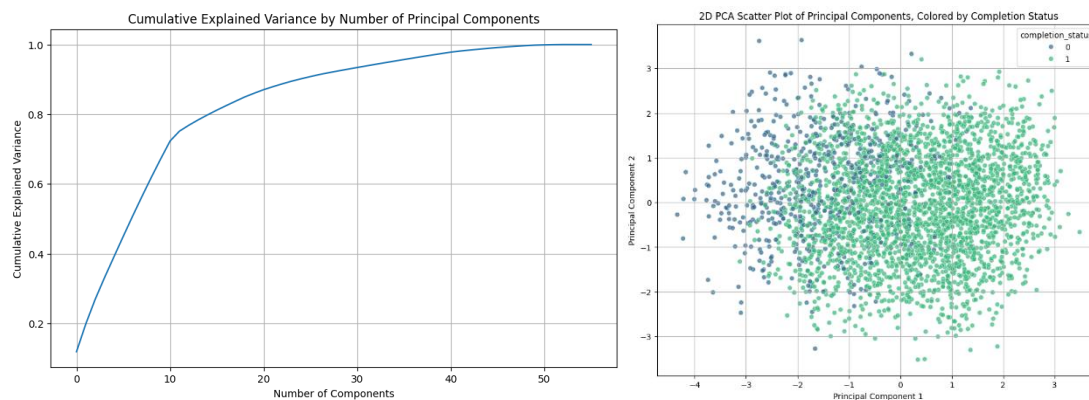
performance, and motivation exhibited the greatest predictive importance, while variables contributing minimal information were excluded from further analysis. Reducing redundant features simplified the model without compromising predictive performance.

The Random Forest Classifier identified several key features influencing student course completion status. The visualization clearly shows the relative importance of each feature.

From the analysis, the following features appear to be most significant:

- **Satisfaction Rating:** This feature has consistently shown a strong correlation with completion status in previous EDA steps, and its high importance confirms its strong predictive power.
- **Engagement Score, Academic Index, Learning Experience Index:** The newly engineered features are highly ranked, indicating their effectiveness in capturing crucial aspects of student behavior and academic performance.
- **Quiz Scores Avg, Assignment Completion Rate, Videos Watched Pct, Hours Per Week:** These individual components of engagement and academic performance also remain highly relevant, contributing significantly to the model's predictive capability.

### 3.7 Principal Component Analysis



**Figure 3.7 Principal Component Analysis**

Principal Component Analysis (PCA) was performed to reduce the dimensionality of the dataset while retaining most of the original information. The scree plot indicates the contribution of each principal component, whereas the cumulative explained variance demonstrates that a relatively small number of components captures the majority of the dataset's variability. PCA reduced computational complexity while preserving the information required for effective prediction.

Principal Component Analysis (PCA) reduces dimensionality by transforming correlated variables into a smaller set of orthogonal principal components while preserving most of the original variance (Jolliffe, 2002).

## CHAPTER 4 MODEL IMPLEMENTATION

### 4.1 Introduction

This chapter presents the implementation, training, and evaluation of the machine learning models developed to predict student course completion in online learning environments. Following the preprocessing and feature engineering procedures described in Chapter 3, multiple supervised learning algorithms were implemented and compared to determine the most suitable model for identifying students who are at risk of dropping out. The models evaluated in this study include Logistic Regression, Decision Tree, Random Forest, Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM). Their predictive performance was assessed using Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Since the primary objective of the study is to identify learners who are likely to drop out, greater emphasis was placed on Recall and F1-Score for the minority class, together with ROC-AUC as an overall measure of classification performance.

### 4.2 Model Development

Five supervised machine learning algorithms were implemented to classify learners into two outcome categories: **Completed** and **Dropped**. The selected algorithms represent different learning approaches, enabling a comprehensive comparison of their predictive capabilities.

Logistic Regression was selected as a baseline linear classifier because of its simplicity and interpretability. Logistic Regression is a widely used statistical classification technique that estimates the probability of binary outcomes through the logistic function (Hosmer et al., 2013). Decision Tree was included for its ability to model nonlinear relationships and produce easily interpretable decision rules. Decision Trees classify observations by recursively partitioning the feature space into homogeneous subsets based on predictor variables (Breiman et al., 1984). Random Forest, an ensemble learning method, was implemented to improve predictive performance by aggregating multiple decision trees. Random Forest improves predictive performance by combining the predictions of multiple decision trees using bootstrap aggregation (Breiman, 2001). XGBoost was selected because of its reputation for achieving high predictive accuracy through gradient boosting techniques. XGBoost employs gradient boosting techniques to iteratively improve prediction accuracy while reducing overfitting through regularization (Chen & Guestrin, 2016). Finally, Support Vector Machine (SVM) was employed due to its effectiveness in handling high-dimensional datasets and complex decision boundaries. Support Vector Machines identify an optimal separating hyperplane that maximizes the margin between classes, making them particularly effective for high-dimensional classification problems (Cortes & Vapnik, 1995).

To improve generalization performance, hyperparameter tuning was performed for the SVM model using Grid Search with cross-validation. In addition, class weighting strategies were incorporated where appropriate to reduce the effect of class imbalance during model training. Hyperparameter optimization using GridSearchCV systematically

evaluates combinations of model parameters to identify configurations that maximize predictive performance (Pedregosa et al., 2011).

### 4.3 Logistic Regression and Data Leakage Assessment

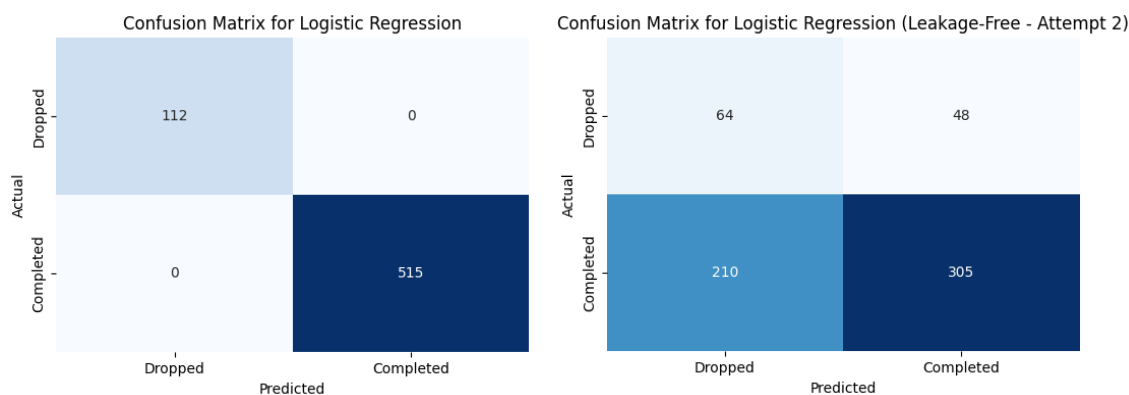
The initial Logistic Regression model was trained using the complete set of predictor variables. During model evaluation, it was observed that variables such as **satisfaction\_rating** and the derived **learning\_experience\_index** contained information that may only become available after a learner completes the course. Including these variables during prediction would result in **target leakage**, causing the model to learn information unavailable during actual deployment.

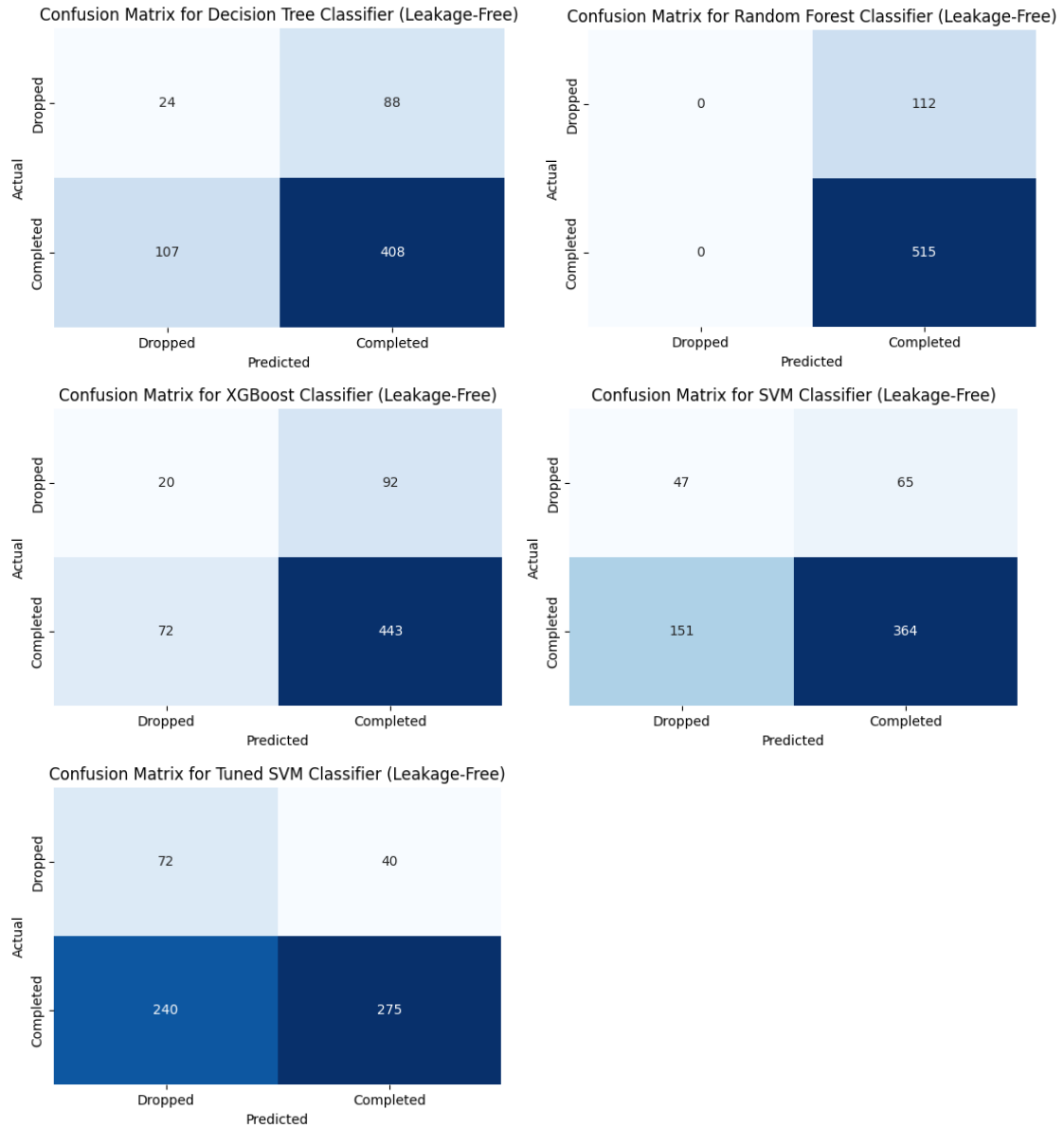
To address this issue, both variables were removed from the predictor set, and the Logistic Regression model was retrained using the leakage-free dataset. This approach ensured that all predictive variables represented information realistically available before the learner's final course outcome was known, thereby improving the validity and applicability of the predictive models.

### 4.4 Model Evaluation

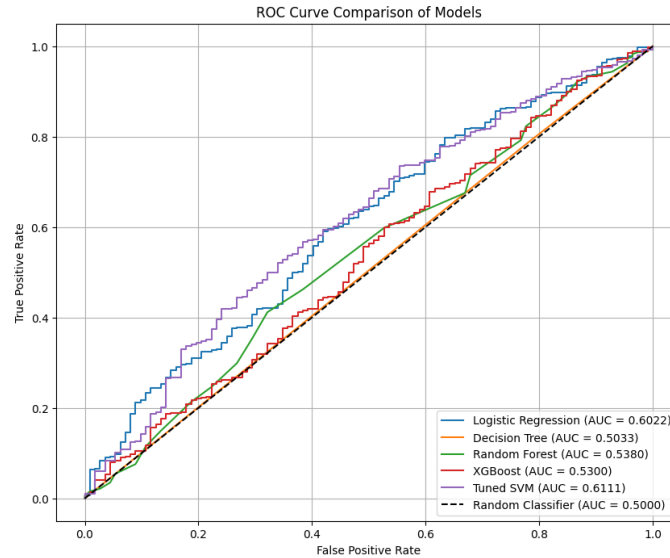
The performance of each machine learning algorithm was evaluated using multiple classification metrics. Accuracy measured the overall proportion of correctly classified learners, while Precision quantified the proportion of predicted dropouts that were correctly identified. Recall measured the model's ability to identify actual dropout cases, making it particularly important for early intervention. The F1-Score provided a balanced assessment of Precision and Recall, whereas ROC-AUC evaluated the model's overall discriminative capability across different classification thresholds.

Confusion matrices, classification reports, and ROC curves were generated for each model to provide a comprehensive evaluation of predictive performance. These visualizations helped identify the strengths and weaknesses of each classifier and supported the selection of the most appropriate model for deployment.





**Figure 4.1 Confusion Matrices**



**Figure 4.2 ROC Curve Comparison of Models**

## 4.5 Comparative Performance of Machine Learning Models

Table 4.1 summarizes the performance of the five machine learning algorithms evaluated in this study.

| Model               | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 0.5885   | 0.23      | 0.57   | 0.33     | 0.6022  |
| Decision Tree       | 0.6890   | 0.18      | 0.21   | 0.20     | 0.5033  |
| Random Forest       | 0.8214   | 0.00      | 0.00   | 0.00     | 0.5380  |
| XGBoost             | 0.7384   | 0.22      | 0.18   | 0.20     | 0.5300  |
| Tuned SVM           | 0.5534   | 0.23      | 0.64   | 0.34     | 0.6111  |

**Table 4.1. Model Performance Comparison**

The results demonstrate notable differences in the predictive performance of the evaluated models. Random Forest achieved the highest overall accuracy (82.14%), indicating strong performance in correctly classifying the majority class. However, it failed to identify any dropout cases, resulting in zero Precision, Recall, and F1-Score for the minority class. This indicates that the model was heavily influenced by class imbalance and is therefore unsuitable for the primary objective of identifying at-risk learners.

Decision Tree and XGBoost achieved moderate classification accuracy but exhibited relatively low Recall and F1-Scores, limiting their effectiveness in detecting learners who were likely to drop out. Logistic Regression demonstrated improved Recall compared with these models but achieved only moderate overall performance.

Among all evaluated algorithms, the **Tuned Support Vector Machine (SVM)** produced the highest Recall (0.64), highest F1-Score (0.34), and highest ROC-AUC (0.6111). Although its overall accuracy was lower than that of Random Forest and XGBoost, the model successfully identified a substantially larger proportion of actual dropout cases. Because the principal objective of this study is to identify learners at risk

of dropping out rather than simply maximize classification accuracy, the Tuned SVM was selected as the final predictive model.

#### **4.6 Best Model Selection**

Model selection was based on the project's primary objective of supporting early intervention for students who are at risk of dropping out. In educational prediction tasks, failing to identify an at-risk learner may prevent timely academic support, making Recall a more appropriate performance measure than overall accuracy.

Based on the comparative evaluation, the Tuned Support Vector Machine demonstrated the best overall balance between identifying dropout cases and maintaining acceptable classification performance. Its superior Recall and ROC-AUC indicate a stronger ability to distinguish between learners who are likely to complete a course and those who are at risk of dropping out. Consequently, the Tuned SVM was selected for the subsequent explainability and fairness analyses presented in Chapter 5.

#### **4.7 Model Reproducibility**

To ensure reproducibility, the final trained Support Vector Machine model and the associated feature scaler were saved using the **Joblib** library. Saving these artifacts allows the trained model to be reloaded without repeating the training process and facilitates future deployment or integration into online learning systems. Model configurations, preprocessing steps, and evaluation procedures were also documented within the project notebook to support transparency and reproducibility.

#### **4.8 Summary**

This chapter presented the implementation and comparative evaluation of five supervised machine learning algorithms for predicting student course completion. While Random Forest achieved the highest overall accuracy, its inability to identify dropout cases limited its practical usefulness. The Tuned Support Vector Machine demonstrated the highest Recall, F1-Score, and ROC-AUC, making it the most appropriate model for identifying learners at risk of dropping out. Accordingly, the Tuned SVM was selected as the final predictive model. The following chapter examines the interpretability, fairness, and ethical implications of the selected model using Explainable Artificial Intelligence (XAI) techniques and bias auditing methods.



## **CHAPTER 5 EXPLAINABLE ARTIFICIAL INTELLIGENCE, ETHICAL AI, AND BIAS AUDITING**

### **5.1 Introduction**

While predictive performance is an essential consideration in machine learning, the practical adoption of artificial intelligence in education also requires transparency, interpretability, and fairness. Educational institutions must understand not only what a predictive model forecasts but also why those predictions are made and whether they are produced equitably across different learner groups. Consequently, this chapter evaluates the selected Support Vector Machine (SVM) model using Explainable Artificial Intelligence (XAI) techniques and fairness auditing methods. These analyses provide insight into the factors influencing learner predictions, assess potential algorithmic bias, and examine the ethical implications of deploying predictive analytics in online learning environments.

### **5.2 Explainable Artificial Intelligence (XAI)**

To improve the transparency of the predictive model, Explainable Artificial Intelligence (XAI) techniques were employed to interpret both global and local model behavior. Specifically, SHapley Additive Explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), Partial Dependence Plots (PDP), and Individual Conditional Expectation (ICE) plots were used to explain how different learner characteristics influenced predictions.

These techniques enable educators and decision-makers to understand the reasoning behind the model's predictions, thereby increasing confidence in the deployment of machine learning for educational decision support.

### **5.3 Global Feature Importance Using SHAP**

SHapley Additive Explanations (SHAP) were used to determine the relative importance of each predictor variable in the selected Support Vector Machine model. The SHAP summary plot illustrates the overall contribution of every feature across all learner predictions and provides a global interpretation of model behavior. SHAP provides both global and local explanations by quantifying the contribution of each feature to individual model predictions using concepts derived from cooperative game theory (Lundberg & Lee, 2017).

The analysis indicates that learner engagement and academic performance variables contribute most significantly to predicting course completion. Variables associated with study habits, academic achievement, learner motivation, and engagement consistently exhibited the largest SHAP values, indicating that they exert the greatest influence on model predictions. In contrast, demographic variables generally contributed less to the prediction process.



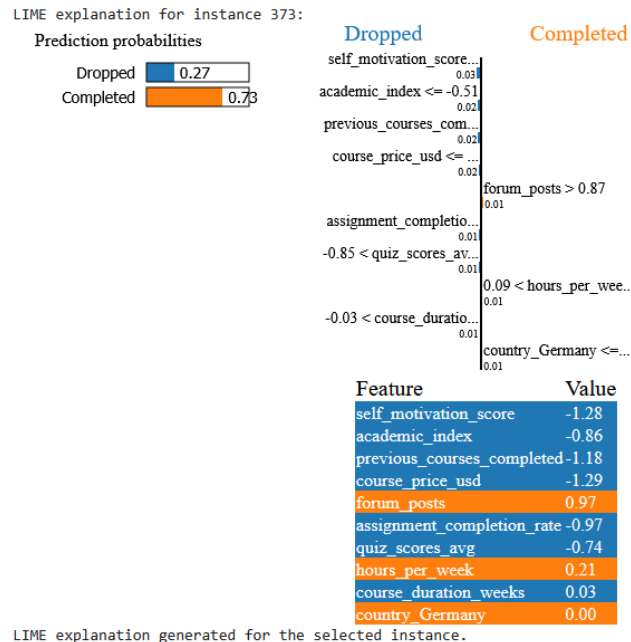
**Figure 5.1 SHAP Summary Plot**

These findings suggest that learner engagement and academic behavior are stronger indicators of course completion than demographic characteristics alone. The SHAP analysis therefore supports the educational validity of the predictive model by demonstrating that predictions are driven primarily by meaningful learning-related factors rather than by personal demographic attributes.

#### 5.4 Local Model Interpretation Using LIME

While SHAP provides a global explanation of model behavior, Local Interpretable Model-Agnostic Explanations (LIME) were employed to explain individual learner predictions. LIME approximates the behavior of the predictive model around a single observation by identifying the variables that most strongly influence a specific prediction. LIME approximates the behavior of the predictive model around a single observation by identifying the variables that most strongly influence a specific prediction.

The LIME explanations demonstrate that the contribution of individual variables varies across learners depending on their unique characteristics and learning behaviors. For example, higher engagement scores, stronger academic performance, and better time management generally increased the predicted probability of course completion, whereas lower engagement and weaker academic indicators contributed to predictions of learner dropout.

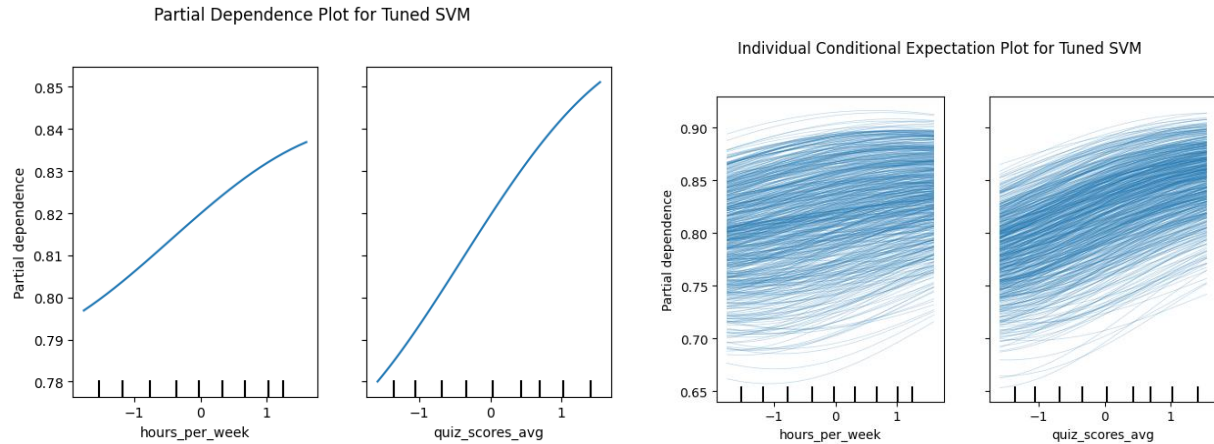


**Figure 5.2 LIME**

The use of LIME enhances the interpretability of individual predictions and enables instructors to understand why a particular learner was identified as being at risk. Such explanations can support personalized academic interventions while improving trust in machine learning-assisted decision making.

## 5.5 Partial Dependence and Individual Conditional Expectation Analysis

To further investigate the influence of important predictor variables, Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots were generated. PDPs illustrate the average effect of a predictor variable on the model's predictions, whereas ICE plots display how changes in a predictor affect individual learner predictions. Partial Dependence Plots visualize the average marginal effect of predictor variables on model predictions, enabling interpretation of complex machine learning models (Friedman, 2001). Individual Conditional Expectation plots complement Partial Dependence Plots by illustrating how changes in predictor variables affect individual observations (Goldstein et al., 2015).



**Figure 5.3 PDP and ICE for Tuned SVM**

The visualizations indicate that improvements in learner engagement and academic performance generally increase the predicted probability of course completion. However, the ICE plots also reveal variability among individual learners, indicating that the impact of specific variables differs across the learner population. This observation highlights the importance of considering multiple learner characteristics simultaneously rather than relying on any single predictor when designing intervention strategies. Together, PDP and ICE analyses demonstrate that the predictive model captures complex and individualized relationships among learner characteristics while remaining interpretable.

## 5.6 Model Limitations

Despite the satisfactory performance of the selected Support Vector Machine model, several limitations should be acknowledged.

First, the dataset exhibited class imbalance, with substantially more learners completing courses than dropping out. Although class weighting and threshold optimization were employed to improve minority class detection, imbalance remains a challenge that may affect predictive performance.

Second, potential target leakage was identified during model development. Variables such as **final\_grade**, **satisfaction\_rating**, and the derived **Learning Experience Index** contain information that may only become available after course completion. These variables were therefore removed from the predictive feature set to ensure that the model relied exclusively on information available prior to predicting learner outcomes.

Finally, the study utilized a synthetic dataset rather than data collected from an operational online learning platform. Consequently, although the developed models demonstrate the feasibility of predictive analytics for educational applications, additional validation using real institutional data is recommended before deployment.

## 5.7 Bias and Fairness Analysis

To promote the responsible use of artificial intelligence, the selected predictive model was evaluated for potential bias across several learner groups, including gender, education level, financial aid status, and employment status. The objective of the fairness analysis was to determine whether the model produced substantially different prediction outcomes for specific demographic or socioeconomic groups. Fairness evaluation seeks to identify and mitigate unintended algorithmic bias that may disproportionately affect specific demographic or socioeconomic groups (Mehrabi et al., 2021; Barocas et al., 2023).

### ### Fairness Analysis: Demographic Parity (Predicted Completion Rate)

#### --- Gender ---

| gender     | predicted_completion |
|------------|----------------------|
| Female     | 0.506329             |
| Male       | 0.485915             |
| Non-binary | 0.62963              |

#### --- Education Level ---

| education_level | predicted_completion |
|-----------------|----------------------|
| Bachelor's      | 0.477366             |
| High School     | 0.549296             |
| Master's        | 0.48951              |
| PhD             | 0.428571             |

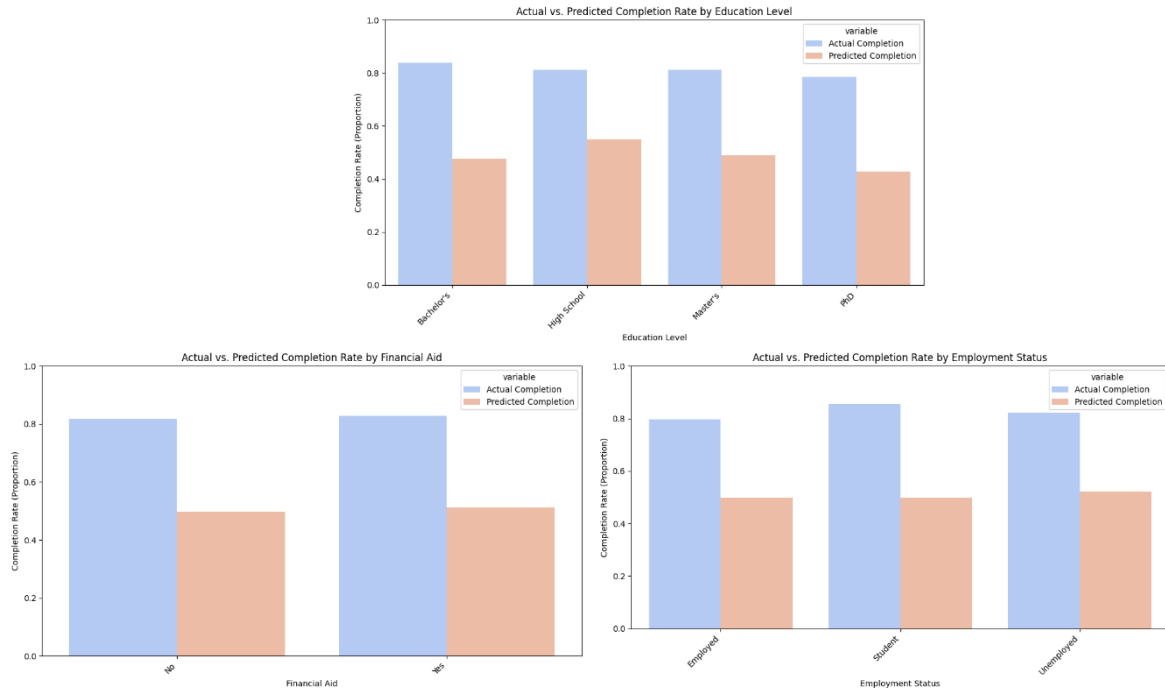
#### --- Financial Aid ---

| financial_aid | predicted_completion |
|---------------|----------------------|
| No            | 0.497573             |
| Yes           | 0.511628             |

#### --- Employment Status ---

| employment_status | predicted_completion |
|-------------------|----------------------|
| Employed          | 0.5                  |
| Student           | 0.5                  |
| Unemployed        | 0.522388             |

**Table 5.1 Fairness Analysis**



**Figure 5.4 Actual vs Predicted**

The fairness assessment compared actual completion rates with predicted completion rates across each subgroup. Overall, the analysis showed that the model produced relatively consistent prediction patterns across the evaluated groups. Although the model tended to slightly under-predict course completion, no evidence of severe systematic bias favoring or disadvantaging any single demographic group was observed.

The findings suggest that the selected Support Vector Machine model demonstrates an acceptable level of fairness while maintaining its ability to identify learners who may require additional academic support.

## 5.8 Fairness Metrics

Several fairness metrics were examined to assess the ethical performance of the predictive model. These included demographic parity, equalized odds, and disparate impact, which evaluate whether prediction outcomes are distributed equitably across different learner groups.

The results indicate that the predictive model exhibits relatively balanced behavior across gender, education level, financial aid status, and employment status. While minor differences in prediction rates were observed, these variations were not sufficiently large to indicate significant algorithmic discrimination. Consequently, the selected model demonstrates reasonable fairness for educational decision-support applications.

Nevertheless, fairness should be regarded as an ongoing evaluation process rather than a one-time assessment. Continuous monitoring is necessary to ensure that future learner populations remain fairly represented as new data become available.

## 5.9 Bias Mitigation

To reduce the likelihood of algorithmic bias, several mitigation strategies were incorporated during model development. Class weighting and decision threshold optimization were employed to improve the detection of minority class instances. Variables identified as potential sources of target leakage were removed from the predictive feature set, thereby preventing unrealistic model performance. Furthermore, fairness analyses were conducted across multiple learner groups to evaluate whether predictions remained equitable.

Future implementations may further enhance fairness through advanced mitigation techniques such as reweighting, data augmentation, post-processing adjustments, or fairness-constrained optimization. These approaches may reduce prediction disparities while maintaining acceptable predictive performance.

## 5.10 Summary

This chapter evaluated the selected Support Vector Machine model from the perspectives of interpretability, fairness, and responsible artificial intelligence. Explainable AI techniques, including SHAP, LIME, Partial Dependence Plots, and Individual Conditional Expectation plots, demonstrated that learner engagement, academic performance, and motivation were the most influential factors driving model predictions. Fairness analyses further indicated that the model produced relatively consistent outcomes across demographic and socioeconomic groups, with no evidence of substantial systematic bias. Although certain limitations remain, the incorporation of explainability and fairness auditing enhances the transparency, accountability, and ethical applicability of the predictive model for supporting student success in online learning environments.

## References

- Baker, R. S., & Inventado, P. S. (2014). *Educational data mining and learning analytics*. In J. A. Larusson & B. White (Eds.), *Learning Analytics* (pp. 61–75). Springer.
- Baker, R. S. J. d., & Yacef, K. (2009). The state of educational data mining in 2009: A review and future visions. *Journal of Educational Data Mining*, 1(1), 3–17.
- Barocas, S., Hardt, M., & Narayanan, A. (2023). *Fairness and Machine Learning*. MIT Press.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and Regression Trees*. Chapman & Hall.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.
- Dhawan, S. (2020). Online learning: A panacea in the time of COVID-19 crisis. *Journal of Educational Technology Systems*, 49(1), 5–22.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.
- Goldstein, A., Kapelner, A., Bleich, J., & Pitkin, E. (2015). Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics*, 24(1), 44–65.
- Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann.
- Hodges, C., Moore, S., Lockee, B., Trust, T., & Bond, A. (2020). The difference between emergency remote teaching and online learning. *EDUCAUSE Review*.
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.
- Jolliffe, I. T. (2002). *Principal Component Analysis* (2nd ed.). Springer.



Kizilcec, R. F., Piech, C., & Schneider, E. (2013). Deconstructing disengagement: Analyzing learner subpopulations in massive open online courses. *Proceedings of the Third International Conference on Learning Analytics and Knowledge*, 170–179.

Kuhn, M., & Johnson, K. (2019). *Feature Engineering and Selection*. CRC Press.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6), 1–35.

Patel, R. (2026). *Online Learning Dataset 2026* [Dataset]. Kaggle.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144.

Romero, C., & Ventura, S. (2020). Educational data mining and learning analytics: An updated survey. *WIREs Data Mining and Knowledge Discovery*, 10(3), e1355.